

lyra-cpp — Remaining Work

What's still open as of **v0.7.0** (2026-06-29). The list keeps shrinking — the big TX bring-up, the native DSP rack, CW transmit + decoder, Profiles, VAC, the panadapter/tuning UI, Spots, the Tuner memory, and the FM transmit refinements are all **shipped and on-air-confirmed**. What's left is mostly two large arcs (RX2/Split and PureSignal) plus a handful of smaller TX and CW follow-ons.

The two big arcs (the bulk of what remains)

- **RX2 + Split** (#96—#101) — second receiver end-to-end:

- DDC1 wire dispatch + a 2nd wdspEngine RxChannel + RX2 freq writer.
- Stereo-split audio routing (RX1 hard-L / RX2 hard-R).
- Focused-RX UI model (2nd VFO LED, focus border, SUB button,

Ctrl+1/Ctrl+2, middle-click swap, per-RX vol/mute).

- SPLIT semantics + per-VFO TX indicators + A→B/B→A/SWAP + per-mode shift-offset.

- Persistence + per-band RX2 freq memory + TCI channel-1 dispatch.
- Extend RIT/XIT to RX2 (was RX1-only when shipped).

- **PureSignal** (the last big TX-port pillar) — predistortion/calibration:

- Prerequisite: TX analyzer port to reference parity (#140 / Stage E.1).
- Then the PS calibration + auto-attenuator + coefficient persistence

arc (Stage G/H).

TX follow-ons (smaller)

- **VOX** (#91) — voice-operated TX: PttSource on the FSM + threshold + open/close hang + anti-VOX.

- **TX Voice Keyer / message memory** (#89).

- **DSP per-mode buffer / filter-size / latency surface** (#159) —

Setup→DSP→Options parity.

- **#114** — post-FrameComposer TX-policy plumbing: panadapter TX offset + PA-enable safety (the ATT-on-TX half already shipped, §15.31/#166).

- **#118** — Step-14 Stage 2b: Wire-LIVE EP6 migration + txworker_
shared-counter + the 7-instrument re-home.

- **#119** — non-HL2 family FrameComposer branches (blocked: needs tester hardware for ANAN / other P1 families).

- **#123** — finish the line-by-line TX reference-fidelity audit (Pass 2 wave 3 was paused).

- **#55** — RX/TX quick-preset chip panel (one-click named-profile recall),
a follow-on to the shipped Profile Manager.

CW follow-ons

- **External CW keyer support** (#171) — Winkeyer/USB: COM-port + keyer-device config surface.

- **CW keyer settings panel** (#186 / C-5) — break-in / hang / PTT-delay /

sidetone settings.

- **CW decoder accuracy — Tier B onward** (#187 next = B1: independent

dah-length tracking + the $-\frac{1}{2}\ln(\text{var})$ term + warm-up boundary fix). The safe A-series shipped in v0.4.10; Tier B/C want a synthetic-CW regression suite + on-air A/B first; Tier D parked.

Audio / routing

- **VAC2** (#103) — second-channel host-audio bridge (e.g. a logger on a device independent of the digital-mode app).

Platform

- **Linux / macOS buildability** — planned but not yet buildable (the Windows-binary DSP engine + win32 wire code). Standing rule: never imply in docs/UI that Linux/macOS runs today.
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Parked / bench-gated (not active work)

- **#156** — restart-after-hard-kill clean reopen (intermittent; parked).
- FM transmit is at a **complete stopping point** as of v0.7.0 (fixes #2, #3a, #3b shipped); any further FM tuning is operator-bench-driven, not a planned arc.

Recently CLOSED (for reference — no longer open)

TX bring-up (SSB/AM/DSB/SAM/FM + native rack) · TX EQ (#50) / RX EQ (#59) · Combinator (#51) · Plate reverb (#52) · TX speech suite (#88) · Profiles (#49) · TX monitor (#90) · SWR protect + drive cap (#169/#170a) · PHROT (#109) · ATT-on-TX (#166) · TX power model + per-band PA calibration + auto-tuning watts cap (v0.6.0) · CW transmit (#105) · CW RX decoder (#173) + A-series accuracy (v0.4.10) · CW macros (#176) · waterfall callsign ID (#175) · CTUN (#174) · RIT/XIT RX1 · VAC1 (#158/#102) · COM-port CAT + PTT (v0.4.11) · Spots subsystem (#182, v0.5.0) · Tuner manual-ATU memory (v0.7.0) · FM transmit refinements (v0.7.0).